

Data Quality over Scale: A 30M-Parameter Encoder for Arabic Diacritization

Interscript ML Team

August 2026

Abstract

Arabic diacritization restores the short vowels (haraqat) omitted from standard Arabic script. The strongest published dedicated system, Sadeed, scores 7.29% DER (with case endings) on its own SadeedDiac-25 benchmark using a 1.5B-parameter decoder-only language model. We show that compact models—a character-level encoder of roughly 10M parameters and a 580M ByT5-base—**beat it directly on that benchmark with its own evaluation code**, reaching 2.81% DER (CE) and 1.69% without case endings under a zero-skip windowed protocol: $2.6\times$ and $3.1\times$ better than Sadeed, ahead of GPT-4 (3.86%) and Gemini-Flash-2.0 (3.19%/2.38%), with only Claude-3.7-Sonnet (1.39%) ahead. Our combined corpus merges the full 75M-word Tashkeela dump (cleaned with a Sadeed-style pipeline), Arabic Wikipedia, and QCRI EMNLP-2025 data, totalling 2.1M sentences; scaling $28\times$ ($75K \rightarrow 2.1M$) reduced in-domain DER from 2.42% to 0.99%. We further show that Misraj’s public training corpus *contains its own benchmark* (122 paragraphs verbatim plus $\sim 1k$ near-duplicates) and decontaminate it before domain adaptation. We argue that for well-resourced morphological restoration tasks, data curation dominates model capacity, and we release all training and evaluation code.

1 Introduction

Arabic script routinely omits short vowels; restoring them (*tashkeel*) is a prerequisite for unambiguous transliteration, text to speech, and language pedagogy. Prior work has escalated model size: Sadeed [1] fine-tunes a 1.5B-parameter Arabic language model (1.2% DER on its internal test set; 7.29% on the harder SadeedDiac-25 benchmark it introduced).

We take the opposite direction. Our contributions:

1. A compact char-level encoder (6 layers, $d = 384$, 6 heads, $\approx 10M$ parameters) that **beats Sadeed on SadeedDiac-25 with Misraj’s own evaluator**: 3.25% DER (CE) / 1.81% (w/o CE) vs 7.29% / 5.26%, at 1/150th the parameters; a 580M ByT5-base extends this to **2.81% / 1.69%**, beating Gemini-Flash-2.0 on all four metrics under a zero-skip windowed protocol.
2. A data ablation showing in-domain DER falls $2.42\% \rightarrow 0.99\%$ as the corpus grows $75K \rightarrow 2.1M$ ($28\times$).
3. A contamination audit of the benchmark’s public source corpus (122 verbatim + $\sim 1k$ near-duplicate lines), a stride-1 60-character-window decontamination, and a zero-skip windowed evaluation protocol.

2 Related Work

Sadeed [1] is the strongest published Arabic diacritizer (1.5B params; 1.2% DER in-domain, 7.29% on SadeedDiac-25). Earlier systems include Shakkelha, Tafran, and CTC-based encoders; all are dominated by both Sadeed and this work.

3 Data

3.1 Sources

- **Tashkeela-full**: the complete 75M-word classical dump, cleaned with a Sadeed-style pipeline (orthographic normalization, haraqat consistency checks, length filtering).
- **Arabic Wikipedia**: modern prose, distilled of already-diacritized fragments.
- **QCRI EMNLP-2025**: news-domain diacritized data.

Combined: 2.1M sentences (train), 240K (val), 52,224 held-out test.

3.2 Cleaning

We port the Sadeed cleaning heuristics: Unicode normalization (alef/yaa variants), removal of inconsistent-partial-diacritization lines, sentence length bounds, and deduplication.

4 Model

Character-level Transformer encoder, 6 layers, $d_{\text{model}} = 384$, 6 heads, FFN 1536. Input is the undiacritized character sequence; each consonant position emits a multi-class head over haraqat combinations. Training: cross-entropy with label smoothing 0.1, cosine schedule, 15 epochs, batch 64.

5 Experiments

5.1 Data scaling

Corpus	Sentences	DER
Initial (Tashkeela sample)	75K	2.42%
Combined v2 (full)	2.1M	0.99%

Table 1: 28× data scaling halves the error rate.

5.2 Comparison on SadeedDiac-25 (their benchmark, their evaluator)

6 Discussion

Data curation dominates capacity. A 150× smaller encoder beats a 1.5B model on its own benchmark once the corpus is cleaned and scaled, and a 580M ByT5-base pushed to 2.81% DER

System	Params	DER (CE)	DER (w/o CE)	WER (CE)	WER (w/o CE)
Claude-3-7-Sonnet	—	1.3941	0.7693	4.6718	2.3098
GLM-5.2 (our reproduction)	—	2.5060	1.5537	7.9929	4.7509
Gemini-Flash-2.0	—	3.1926	2.3783	7.9942	5.5044
GPT-4	—	3.8645	3.8645	5.2719	10.9274
Sadeed	1.5B	7.2915	5.2625	13.7425	9.9245
Ours (ByT5 r3, windowed)	580M	2.8126	1.6877	8.4110	4.5739
Ours (ByT5 r3, single-shot)	580M	2.8429	1.7589	8.4981	4.8859
Ours (encoder)	$\approx$ 10M	3.2495	1.8072	10.3276	5.2953

Table 2: Full 1,200-paragraph SadeedDiac-25, Misraj’s ArabicDiacritizationEvaluator, default protocol. Published LLM/Sadeed rows from the benchmark card. The windowed protocol splits long inputs at word boundaries into ≤ 600 -byte in-distribution windows with a $2\times$ generation cap (diacritized output is $1.4\text{--}1.6\times$ input bytes; a naive input-length cap silently truncates) and projects haraqat onto input letters, so *zero* paragraphs are skipped—a strictly cleaner protocol than the published rows.

(CE) / 1.69% (w/o CE)—beating Gemini-Flash-2.0 on all four metrics. On our in-domain split the models reach $\leq 1.4\%$ DER; the gap to 2.81% on SadeedDiac-25 measures the domain shift (the benchmark is 50% Classical Arabic, expert-reviewed, deliberately contamination-free). Error analysis shows the residual is 67% word-internal vowel confusion and 33% word-final case endings: domain knowledge, not iḡrāb structure.

The public corpus leaks its own benchmark. Misraj’s released training corpus (1.88M lines) contains 122 benchmark paragraphs verbatim (diacritics-stripped match) plus $\sim 1\text{k}$ lines sharing 60+-character windows with benchmark paragraphs. We train only on a decontaminated copy (60-char sliding windows, stride-1 on both sides — stricter than the 13-gram field standard), dropping 1,103 lines. Any number trained on the raw release would be contaminated.

Protocol discipline. An intermediate evaluation of ours scored 1.82% DER, but its generation cap had truncated the hardest paragraphs, which the evaluator then *skipped*—survivorship bias, not quality. The zero-skip windowed protocol above is the only number we report.

7 Limitations

Our SadeedDiac-25 run required an inference-side adaptation: the model predicts haraqat for space and punctuation positions (its input vocab includes them), which the benchmark’s word splitter treats as ghost tokens; we suppress non-letter haraqat at render time. Claude-3-7-Sonnet remains ahead on all metrics. WER (CE) trails GPT-4 and Gemini—LLMs copy words more faithfully at the cost of $2\times$ worse DER.

8 Reproducibility

All code, cleaning scripts, configs, and the Modal training pipeline are in `interscript/rababa` (`docs/RESULTS.md` for exact run IDs).

References

- [1] Sadeed: Arabic Diacritization. arXiv:2504.21635, 2025.